one in which the universe literally rips itself apart. Next is vacuum decay, the spontaneous production of a *quantum bubble of death*[fn9] that devours the cosmos. Finally, we'll venture into the speculative territory of cyclic cosmology, including theories with extra dimensions of space, in which our cosmos might be obliterated by a collision with a parallel universe … over and over again. The closing chapter will bring it all together with an update from several experts currently working on the cutting edge on which scenario looks most plausible now, and what we can expect to learn from new telescopes and experiments to settle the question once and for all.

What that means for us as human beings, living our little lives in all this inconsiderate vastness, is another question entirely. We'll present a range of perspectives in the epilogue, and address whether or not sentience itself could have any kind of legacy that endures beyond our destruction.[fn10]

We don't know yet whether the universe will end in fire, ice, or something altogether more outlandish. What we do know is that it's an immense, beautiful, truly awesome place, and it's well worth our time to go out of our way to explore it. While we still can.